

Lebin LIANG

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 [LebinLiang](#) |  [LebinLiang](#)

Shenzhen, Guangdong, China

OBJECTIVE

Actively seeking PhD positions for Fall 2026 enrollment or later, with a dedicated research focus on Human-Computer Interaction (HCI) and Human-Robot Interaction (HRI).

EXPERIENCE

- **SONY (CHINA) CO., LTD SHENZHEN BRANCH (RDC-Shenzhen)**  Jan 2025 - Aug 2025
Robotics Inter
Shenzhen, China
- **DJI (RoboMaster)**  Summer/Winter 2020 - 2023
Teaching Assistant
Shenzhen, China

EDUCATION

- **University of Chinese Academy of Sciences (U.S. News 54)** Sept 2023 - June 2026
M.Eng. in Electronic Information
Beijing, China
 - GPA: 3.77/4.0
 - Advisors: [Prof. Can Wang](#) and [Prof. Xinyu Wu](#), Shenzhen Institute of Advanced Technology (SIAT), CAS
- **South China Agricultural University (Best Chinese Universities Ranking 100)** Sept 2019 - June 2023
B.Eng. in Agricultural Mechanization and Automation
Guangzhou, China
 - GPA: 4.31/5.0 (Rank: 1st/30)
 - Advisor: [Prof. Zhiyan Zhou](#) and [Prof. Xiwen Luo](#), South China Agricultural University and Academician of the Chinese Academy of Engineering

HONORS AND AWARDS

- **National Scholarship for Undergraduates** Oct. 2020
Awarded by Ministry of Education of China
 - Prestigious national-level scholarship, awarded to top 1% of undergraduates in China.
- **First-Class Scholarship** 2021 - 2022
South China Agricultural University
 - Merit-based academic scholarship, awarded annually to top-performing students at SCAU.
- **Academic Excellence Scholarship** 2023
Awarded by University of Chinese Academy of Sciences
- **IEEE ICRA 2024 RoboMaster University Sim2Real Challenge** June 2024 
Awarded by IEEE International Conference on Robotics and Automation
 - Achieved **Second Prize** in a competitive university-level Sim2Real robotics challenge.
- **21st National Collegiate RoboMaster Competition - Super Confrontation** Aug 2022 
Organized by DJI
 - Awarded **National First Prize** as a key member of the university team.
- **16th National University Smart Car Competition - iFlytek Smart Restaurant Group** Aug 2021 
Organized by Ministry of Education, China
 - Awarded **National First Prize**, demonstrating excellence in autonomous systems and task completion.

PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [C1] **Lebin Liang**, Haotian Rao, Guohao Shen. (2023). **A Real-time Framework for UAV Indoor Self-Positioning and 3D Mapping Base on 2D Lidar, Stereo Camera and IMU**. In *Proceedings of the IEEE International Conference on Real-time Computing and Robotics (RCAR 2023)*, pp. Datong, China. DOI: 10.1109/RCAR58764.2023.10249971. (First Author)
- [C2] **Lebin Liang**, et al. (2025). **A Biomimetic 4WS-4WD Robotic Platform with Single-Neuron PID Control for Autonomous Field Navigation**. *Proceedings of the 2025 IEEE International Conference on Robotics and Biomimetics (ROBIO)*, Chengdu, China, pp. 829-835. DOI: 10.1109/ROBIO66223.2025.11377803. (First Author)

PROJECTS

• Remember-Together: Multi-Robot Collaborative VLN Framework

2024 - 2025

First Author

[▶ Video](#)

- Proposed a distributed multi-robot framework for constructing long-term collaborative memory to enhance vision-language navigation (VLN) in complex indoor environments.
- Designed a probabilistic spatio-temporal consistency verification algorithm to suppress hallucinations from vision-language models (VLMs) via confidence- and similarity-based filtering.
- Developed a spatial-semantic memory graph structure that organizes episodic data into object-place-event nodes for efficient retrieval and reasoning.

• Legged Robot Control System

2025.02-08

SONY RDC Robotics Project

[▶ Video](#)

- Developed STM32-based low-level control for multi-joint legged robots, including custom PCB driver boards with CAN, RS485, and EtherCAT. Built dynamics models and RL-based motion control with Sim2Real validation in Isaac Gym. Implemented safety features such as disconnection detection, emergency stop, and power monitoring.

• 4WD-4WS Autonomous Sprayer Navigation Control System

2021 - 2023

Project Lead, National Student Innovation Program

[▶ Video](#)

- Led development of a 4WD-4WS agricultural sprayer with RTK-GPS & IMU for autonomous row shifting and omnidirectional path tracking.
- Built an 8-DOF chassis kinematic model with PID trajectory tracking, and designed an STM32-based embedded controller with CAN/USB communication.
- Outcome: Project rated "Excellent," with 1 software copyright and 1 co-authored EI-indexed paper.

• Indoor Autonomous Quadrotor UAV System with RGBD & LiDAR Fusion

2022 - 2023

Individual Project

[▶ Video](#)

- Developed VIO-based indoor UAV localization using RGBD cameras, achieving precise position control.
- Integrated PX4 hardware, LiDAR navigation, and 3D mapping within a ROS framework; evaluated VIO algorithms (VINS-Mono, ORB-SLAM2).
- Built LiDAR point cloud transformations and occupancy grid maps to support autonomous navigation.

• Autonomous Infantry Robot Development (RoboMaster Competition)

2021 - 2022

Navigation Team Lead, Taurus Robotics Team (SCAU)

[▶ Video](#)

- Led a team in developing navigation and perception systems for RoboMaster AI Challenge infantry robots.
- Built ROS-based architecture with Gazebo simulation; implemented sensor fusion (odometry, IMU, LiDAR) for localization and navigation.
- Applied computer vision (OpenCV) for armor recognition and autonomous target tracking; managed STM32 hardware and software integration.

• National University Smart Car Competition

2020 - 2021

Team Captain (2021) / Member (2020)

[▶ Video](#)

- **As Captain (2021, iFlytek Smart Restaurant Group):** Led team to develop a Mecanum wheel robot; implemented indoor localization and multi-task allocation (QR code recognition, target detection) using LiDAR & IMU; built ROS-based multi-task scheduling framework. (National First Prize)
- **As Member (2020, Outdoor Opto-Electronic Group):** Contributed to Ackermann steering car for indoor localization & path planning; designed its embedded controller.

SKILLS

- **Languages:** English (CET-6), Mandarin (Native), Cantonese (Native)
- **Programming:** C++, Python, C, Shell Scripting
- **Technologies & Software:** ROS (Noetic, Humble), Linux (Ubuntu), Git, Docker, STM32 Microcontrollers, FreeRTOS, ESP32, TensorFlow, PyTorch (Basic)
- **Robotics & AI:** SLAM, Visual-Inertial Odometry (VIO), Path Planning, Computer Vision, Control Systems, RL (Basic), Sensor Fusion (LiDAR, IMU, Camera)
- **CAD/EDA Tools:** Altium Designer, SolidWorks, MATLAB, Gazebo, Isaac Gym

ADDITIONAL INFORMATION

- **Interests:** Robotics (Legged, Mobile, UAVs), AI for Robotics, Embedded Systems Design, Autonomous Navigation, Computer Vision, Control Systems, Human-Computer Interaction and Human-Robot Interaction
- **Portfolio:** LebinLiang.github.io
- **Google Scholar:** [Lebin Liang](#)
- **Availability for PhD:** Fall 2026 or later.